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Program 4: Integrate Git with Docker for Source Controlled Application Builds

Project structure:

```
.
├── Dockerfile
├── index.js
├── package.json
├── node_modules
│   └── .....
└── package-lock.json
```

STEP 1: Create a Dockerfile and add the following content.

```
FROM node:latest
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["npm", "start"]
```

STEP 1.2: Create a index.js and add the following content.

```
import express from "express";
const app = express();
const PORT = 3000;
app.use(express.json());
app.get("/", (req, res) => {
  res.send("docker is runnig");
});
app.listen(PORT,"0.0.0.0", () => console.log(`Server running on port ${PORT}`));
```

STEP 1.2: Create a node project

```
npm init -y
npm install express
```

STEP 2: Execute the Docker Build command on the Terminal to build the Docker image and then execute the Docker run command specifying the Port Number for the container.

➤ **sudo docker build -t node1 .**

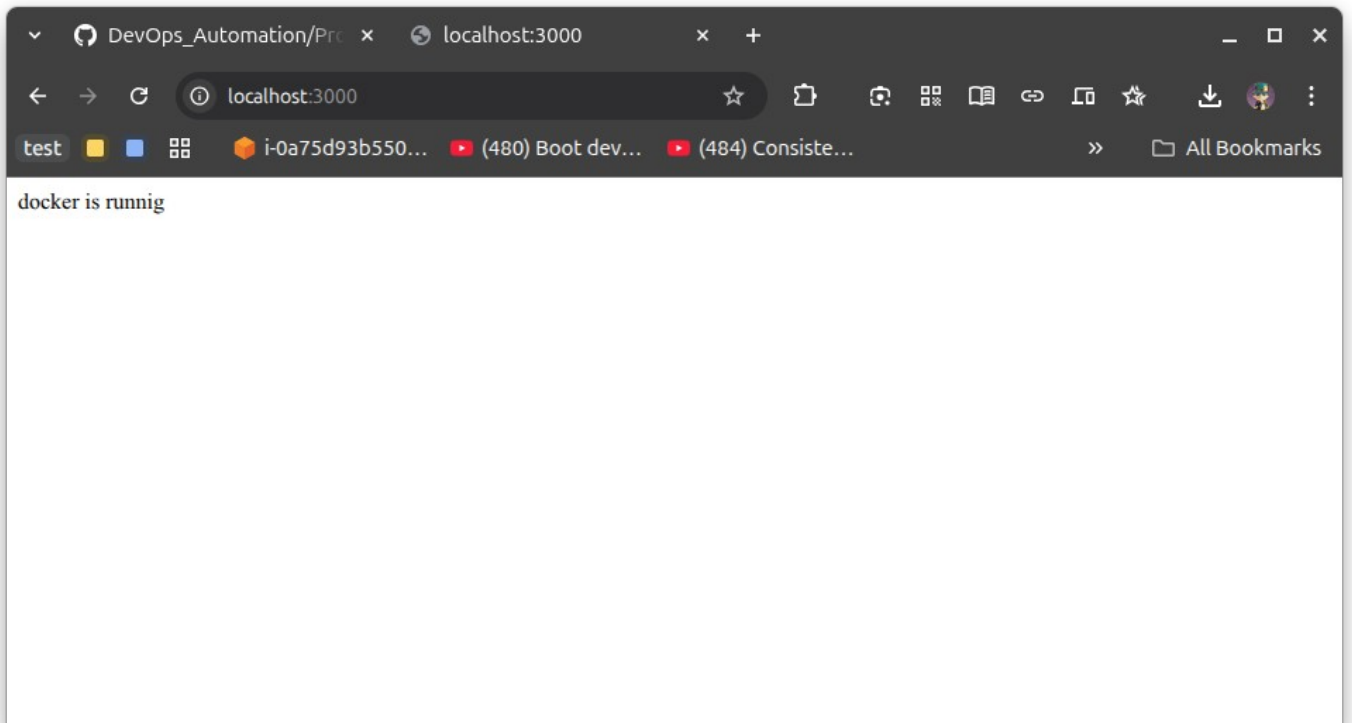
```
> sudo docker build -t node1 .
[sudo] password for kumars:
Sorry, try again.
[sudo] password for kumars:
[+] Building 18.8s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 153B                                0.0s
=> [internal] load metadata for docker.io/library/node:latest     4.2s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:latest@sha256:a84dedcbbcf651db49178dd876bad1ea 13.3s
=> => resolve docker.io/library/node:latest@sha256:a84dedcbbcf651db49178dd876bad1ea6 0.0s
=> => sha256:876cf39ddf32fd345a43c42d3dc00bfe59ab966bbcb3f957a98b90 6.74kB / 6.74kB 0.0s
```

STEP 3: run the containers.

```
> sudo docker images
INFO Info → U In Use
IMAGE          ID                DISK USAGE  CONTENT SIZE  EXTRA
node1:latest    9cf1a6b06e51      1.14GB      0B
> sudo docker run -p 3000:3000 node1

> docker@1.0.0 start
> node index.js

Server running on port 3000
█
```



STEP 4: Stop and remove the container with the container number, remove the docker container and delete the Docker Image

› **sudo docker rmi node1**

STEP 5: in project main folder where the package.json resides execute these commands
Initialize Git Repository locally

```
› git init
› touch .gitignore
› echo "node*/" > .gitignore
› git add .gitignore
› git commit -m "ignore file"
› git add .
› git commit -m "added all files"
› git branch -m -f main
```

STEP 6: Create a New Repo on Github

STEP 7: Get a personal access token (PAT)

Go to profile -> Developer settings -> Personal Access Token -> Token(classic) and generate a token.

STEP 8: Link to Remote Repository

➤ **git remote add origin https://<PAT>@github.com/<username>/<repo-name>.git**

STEP 9: Push code to Github

➤ **git push -u origin main**

